Jarzynski equality for a pulled polymer in 2D

Model, numerical scheme and results

6 October 2026

Abstract

A two-dimensional bead–spring chain is anchored at one end and pulled at the other by a harmonic trap moving at constant speed through a Langevin heat bath. From 10^5 independent pulls per parameter set we measure the distribution of the work W and estimate the free-energy difference ΔF with the Jarzynski equality. Every estimate is compared with an exact value: a closed form for the Gaussian chain, and a log-space convolution for the stiff-bond chain. We vary the pulling time τ, the temperature T and the number of beads N. The equality holds in all cases where enough pulls are available. What decides whether it is usable is the dissipated work in units of $k_{\mathrm{B}}T$, βW_{diss}. This quantity is set by friction and pulling speed, and an affine-stretching argument predicts it within 12% for $N \geq 10$. At fixed τ/τ_1 it is independent of temperature but grows linearly with N, so the number of pulls needed grows exponentially with chain length. Reverse pulls confirm the Crooks fluctuation theorem, and the two-sided Bennett estimator recovers ΔF where one-sided Jarzynski fails.

Full derivations (the three proofs of the Jarzynski equality, how it determines F, the Crooks theorem, and the derivation of every estimator) are in the companion **theory document** `theory.pdf`. This report summarises them and focuses on the results.

Main results

1. **Validation.** For the Gaussian chain the simulated $\langle W \rangle$ and $\operatorname{Var} W$ agree with the exact moments of the discrete algorithm to 0.1% and 1%. The discrete scheme satisfies Jarzynski to $O(\Delta t^2)$ ($8 \times 10^{-4}\, k_\mathrm{B}T$ at $\Delta t = 0.01$).
2. **Entropic vs. energetic elasticity.** For the stiff-bond chain $\Delta F \propto T$ (rubber elasticity); for the Gaussian chain ΔF is independent of T. At fixed τ/τ_1 the work distributions of the stiff-bond chain at different temperatures collapse onto one curve in the variable βW.
3. **Dissipation law.** For slow enough pulls, $\beta W_\mathrm{diss} \simeq \pi^2 N/(6\,\tau/\tau_1)$: it depends only on the reduced speed and on N, not on T. At fixed τ, $\beta W_\mathrm{diss} \propto 1/T$.
4. **Convergence.** $\widehat{\Delta F}_J$ is accurate when $M \gg e^{\beta W_\mathrm{diss}}$. In the intermediate regime its typical error is upward, but a single rare trajectory can push it below ΔF. Bootstrap error bars become over-confident by about 3×.
5. **Size.** At fixed τ/τ_1 and fixed relative extension, βW_diss grows linearly with N ($\beta W_\mathrm{diss} = 16$ at $N = 40$), so the required number of pulls grows exponentially with chain length. This is why Jarzynski is a single-molecule tool.
6. **Where the average comes from.** Plots of $e^{-\beta W}$ show that the exponential average is carried by pulls typical of the *reverse* process. For fast pulls or long chains no forward pull reaches that region, and the estimate silently misses up to half of $\langle e^{-\beta W} \rangle$.
7. **Crooks and BAR.** Reverse pulls confirm the Crooks theorem (log-ratio slope 0.98–0.99, theory 1). Combining both directions with Bennett's acceptance ratio recovers ΔF to within $0.01\, k_\mathrm{B}T$ at $N = 40$ (19.688 ± 0.039 vs. 19.694), where forward and reverse Jarzynski are off by $+1.47$ and $-0.76\, k_\mathrm{B}T$.

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1 Physical model

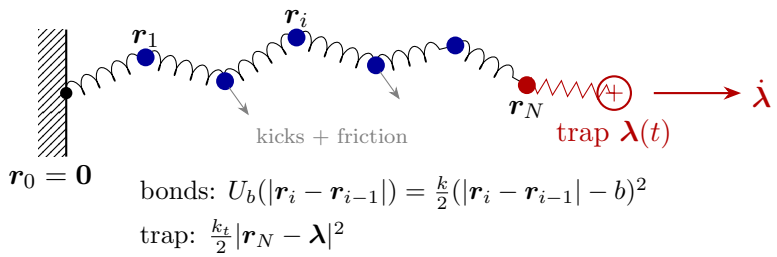


Figure 1: The system. Bead 0 is fixed at the origin. Beads $1 \dots N$ move in the plane and exchange energy with the bath. Bead N is held by a harmonic trap whose centre moves from $\boldsymbol{\lambda}_0$ to $\boldsymbol{\lambda}_1$ in a time τ.

The state is $\boldsymbol{x} = (\boldsymbol{r}_1, \dots, \boldsymbol{r}_N; \boldsymbol{p}_1, \dots, \boldsymbol{p}_N) \in \mathbb{R}^{4N}$, with $\boldsymbol{r}_0 \equiv \mathbf{0}$. The Hamiltonian depends on the trap position $\boldsymbol{\lambda} \in \mathbb{R}^2$:

$$H(\boldsymbol{x}; \boldsymbol{\lambda}) = \sum_{i=1}^{N} \frac{|\boldsymbol{p}_i|^2}{2m} + \sum_{i=1}^{N} \frac{k}{2}\big(|\boldsymbol{r}_i - \boldsymbol{r}_{i-1}| - b\big)^2 + \frac{k_t}{2}\,|\boldsymbol{r}_N - \boldsymbol{\lambda}|^2. \tag{1}$$

N is the number of mobile beads, which equals the number of bonds. The bond rest length b selects one of two kinds of elasticity:

- $b = 0$, **Gaussian (Rouse) chain**. All forces are linear, so ΔF and even the full work distribution are known exactly. The elasticity is *energetic*: stretching stores energy in the springs. This chain is used to validate the code.
- $b = 1$, $kb^2 \gg k_{\mathrm{B}}T$, **nearly freely-jointed chain**. Bond lengths barely change, so stretching works against *entropy*, because a stretched chain has fewer available shapes. This is the physics model.

Simplifications: the chain is a phantom chain (no excluded volume, so it can cross itself), it has no bending stiffness, and the bath is free-draining (no hydrodynamic interactions between beads).

Protocol. The trap moves along x at constant speed, $\boldsymbol{\lambda}(t) = (x_0 + (x_1 - x_0)t/\tau, 0)$, from $x_0 = 0.2L$ to $x_1 = 0.7L$, where $L = Nb$ is the contour length. For $b = 0$ we use the nominal length $L = 0.3N$. The chain starts in equilibrium at $\boldsymbol{\lambda}_0$; at $t = \tau$ it is not in equilibrium.

Units and parameters. We use $m = b = \varepsilon = 1$ (time unit $\sqrt{mb^2/\varepsilon}$), bond stiffness $k = 100$, trap stiffness $k_t = 10$, friction rate $\gamma = 1$ (bead friction coefficient $\zeta = m\gamma$), time step $\Delta t = 0.01$, and $M = 10^5$ independent pulls per parameter set. Sweeps: $k_{\mathrm{B}}T \in \{0.25, 0.5, 1, 2\}$ and $N \in \{5, 10, 20, 40\}$.

2 Heat bath: Langevin dynamics

Each bead obeys

$$\mathrm{d}\boldsymbol{r}_i = \frac{\boldsymbol{p}_i}{m}\mathrm{d}t, \qquad \mathrm{d}\boldsymbol{p}_i = -\nabla_{\boldsymbol{r}_i} H\,\mathrm{d}t - \gamma\,\boldsymbol{p}_i\,\mathrm{d}t + \sqrt{2\gamma m k_{\mathrm{B}}T}\;\mathrm{d}\boldsymbol{B}_i, \tag{2}$$

with independent 2D Wiener processes $\boldsymbol{B}_i$. Solvent collisions enter in two forms, a systematic drag $-\gamma\boldsymbol{p}$ and random kicks. Random kicks alone would heat the chain without bound: for a free bead with noise strength σ, $\frac{\mathrm{d}}{\mathrm{d}t}\langle p^2\rangle = -2\gamma\langle p^2\rangle + \sigma^2$, which grows linearly in time when $\gamma = 0$. With friction, equipartition $\langle p^2\rangle = mk_{\mathrm{B}}T$ fixes $\sigma^2 = 2\gamma m k_{\mathrm{B}}T$. This is the fluctuation–dissipation relation. At frozen $\boldsymbol{\lambda}$ the Kramers equation of (2) has the canonical distribution $\rho_{\mathrm{eq}} = e^{-\beta H}/Z(\boldsymbol{\lambda})$ as its stationary solution, and the free energy is $F(\boldsymbol{\lambda}) = -k_{\mathrm{B}}T \ln Z(\boldsymbol{\lambda})$.

Slowest relaxation time. A pull is "fast" or "slow" only relative to how quickly the chain re-equilibrates, so every pulling time is quoted in units of τ_1, the relaxation time of the slowest collective motion of the chain (derivation in the theory document, Sec. 3.2). It is obtained in four steps:

1. On chain scales inertia is negligible (momenta relax in $1/\gamma = 1$), so beads move overdamped, $\zeta \dot{\boldsymbol{r}}_i = \boldsymbol{F}_i + \text{noise}$.
2. Each bond is replaced by a linear spring κ: $\kappa = k$ for $b = 0$. For $b = 1$ the bonds do not stretch, but by the central limit theorem a chain segment has a Gaussian end-to-end distribution, i.e. it acts as an *entropic* spring with $\kappa = 2k_{\mathrm{B}}T/\langle u^2 \rangle \approx 1.94$ per bond in 2D.
3. The linear chain (bead 0 fixed, trap k_t on bead N) has stiffness matrix K, and each normal mode relaxes as $e^{-\lambda_p t/\zeta}$. The slowest gives $\tau_1 = \zeta/\lambda_{\min}(K)$, computed numerically.
4. Since $k_t = 10 \gg \kappa$, the trap almost clamps the far end. The slowest mode is then the half-wave $\sin(\pi i/N)$ of a string fixed at both ends, with $\lambda_{\min} \simeq \kappa\pi^2/N^2$, and

$$\tau_1 \simeq \frac{\zeta N^2 \langle u^2 \rangle}{2\pi^2 k_{\mathrm{B}}T} \qquad (b = 1). \tag{3}$$

(5.22 for $N = 10$, against the exact eigenvalue 5.46). $\tau_1 \propto N^2$ is the Rouse scaling: a disturbance must diffuse along the whole chain. $\tau_1 \propto 1/T$ for the entropic chain, because the entropic spring stiffens with temperature. *Correction to the theory draft:* the draft assumed a free far end, i.e. a quarter-wave mode, which overestimates τ_1 by a factor 4. We use the numerically computed eigenvalue throughout: $\tau_1 = 1.45, 5.46, 21.3, 84.3$ for $N = 5, 10, 20, 40$ at $k_{\mathrm{B}}T = 1$. Since $\tau_1 \propto 1/T$, $\tau_1 = 20.8$ at $k_{\mathrm{B}}T = 0.25$ and 2.91 at $k_{\mathrm{B}}T = 2$ ($N = 10$). For the Gaussian chain, $\tau_1 = 0.26$ ($N = 10$), independent of T.

3 Stochastic thermodynamics and the Jarzynski equality

This section summarises the theory. Step-by-step derivations are in the theory document (`theory.pdf`): Sec. 4 (free energy and reversible work), Sec. 5 (three proofs of the Jarzynski equality), Sec. 6 (what the equality says about F), Sec. 7 (Crooks) and Sec. 8 (estimators).

Work. The work done by the puller along one trajectory is

$$W = \int_0^\tau \frac{\partial H}{\partial \boldsymbol{\lambda}} \cdot \dot{\boldsymbol{\lambda}} \, \mathrm{d}t = -k_t \int_0^\tau (\boldsymbol{r}_N - \boldsymbol{\lambda}) \cdot \dot{\boldsymbol{\lambda}} \, \mathrm{d}t, \tag{4}$$

and the heat absorbed from the bath is $Q = \Delta H - W$. The second law gives $\langle W \rangle \geq \Delta F$, and $W_{\mathrm{diss}} = \langle W \rangle - \Delta F$ is the dissipated work.

Jarzynski equality. If the chain starts in equilibrium at $\boldsymbol{\lambda}_0$ and the dynamics at every frozen $\boldsymbol{\lambda}$ preserves $\rho_{\mathrm{eq}}(\cdot; \boldsymbol{\lambda})$, then for any protocol

$$\left\langle e^{-\beta W} \right\rangle = e^{-\beta \Delta F}. \tag{5}$$

Discrete-time proof, which is what the code implements. Split the protocol into steps $\boldsymbol{\lambda}_0, \ldots, \boldsymbol{\lambda}_K$ and alternate two moves. First a *switch* $\boldsymbol{\lambda}_k \to \boldsymbol{\lambda}_{k+1}$ at frozen $\boldsymbol{x}$, which costs $\delta W_k = H(\boldsymbol{x}_k; \boldsymbol{\lambda}_{k+1}) - H(\boldsymbol{x}_k; \boldsymbol{\lambda}_k)$. Then a *relax* move $\boldsymbol{x}_k \to \boldsymbol{x}_{k+1}$ by a Markov kernel P_{k+1} that preserves $\rho_{\mathrm{eq}}(\cdot; \boldsymbol{\lambda}_{k+1})$. Define $f_k(\boldsymbol{y}) = \left\langle \delta(\boldsymbol{x}_k - \boldsymbol{y}) e^{-\beta W_k} \right\rangle$. Then $f_0 = e^{-\beta H(\cdot; \boldsymbol{\lambda}_0)}/Z_0$ and

$$f_{k+1}(\boldsymbol{y}) = \int f_k(\boldsymbol{x}) \, e^{-\beta \delta W_k} P_{k+1}(\boldsymbol{x} \to \boldsymbol{y}) \, \mathrm{d}\boldsymbol{x} = \frac{1}{Z_0} \int e^{-\beta H(\boldsymbol{x}; \boldsymbol{\lambda}_{k+1})} P_{k+1}(\boldsymbol{x} \to \boldsymbol{y}) \, \mathrm{d}\boldsymbol{x} = \frac{e^{-\beta H(\boldsymbol{y}; \boldsymbol{\lambda}_{k+1})}}{Z_0}, \tag{6}$$

so $\langle e^{-\beta W}\rangle = \int f_K = Z_K/Z_0$. The kinetic energy does not depend on $\boldsymbol{\lambda}$ and drops out of ΔF. Any discretised Langevin step preserves ρ_{eq} only approximately, so the equality holds up to the integrator's invariant-measure error (Sec. 5.1).

Cumulants. Expanding in cumulants, $\Delta F = \langle W\rangle - \frac{\beta}{2}\sigma_W^2 + \frac{\beta^2}{6}\kappa_3 - \dots$. When W is Gaussian (exactly true for $b = 0$, where W is a linear functional of a Gaussian process), $W_{\mathrm{diss}} = \beta\sigma_W^2/2$. Near equilibrium the same relation $\sigma_W^2 \simeq 2k_{\mathrm{B}}TW_{\mathrm{diss}}$ holds for any system (linear response).

From the equality to F. Taking the logarithm, $\Delta F = -k_{\mathrm{B}}T\ln\langle e^{-\beta W}\rangle$: nonequilibrium work measurements give an *equilibrium* free-energy difference. Only differences of F are accessible, and in the quasi-static limit $W \to \Delta F$ because $\partial_{\boldsymbol{\lambda}}F = \langle\partial_{\boldsymbol{\lambda}}H\rangle_{\boldsymbol{\lambda}}$. Jensen's inequality gives the second law, and $\Pr(W < \Delta F - \zeta) \le e^{-\beta\zeta}$: the pulls that dominate the average are exponentially rare.

Crooks theorem and two-sided estimators. With P_R the work distribution of the reversed protocol ($\boldsymbol{\lambda}_1 \to \boldsymbol{\lambda}_0$, started in equilibrium at $\boldsymbol{\lambda}_1$),

$$\frac{P_F(W)}{P_R(-W)} = e^{\beta(W-\Delta F)}, \qquad \text{equivalently} \qquad \frac{e^{-\beta W}P_F(W)}{\langle e^{-\beta W}\rangle_F} = P_R(-W). \tag{7}$$

The second form says that the share of the exponential average contributed by forward works near W *is* the mirrored reverse distribution. Bennett's acceptance ratio (BAR) solves $\sum_F \mathrm{f}(\beta(W_m - \widehat{\Delta F}) + \ln\frac{M}{M_R}) = \sum_R \mathrm{f}(\beta(W_m^R + \widehat{\Delta F}) - \ln\frac{M}{M_R})$, $\mathrm{f}(x) = 1/(1+e^x)$. It is the minimum-variance (and maximum-likelihood) way to combine forward and reverse pulls.

How many pulls are needed. The estimator $\widehat{\Delta F}_J = -k_{\mathrm{B}}T\ln\big(M^{-1}\sum_m e^{-\beta W_m}\big)$ is dominated by rare pulls with $W < \Delta F$. By (7), these are the pulls typical of the reverse process, near $W \approx \Delta F - W_{\mathrm{diss}}^R$, and they have probability $\sim e^{-\beta W_{\mathrm{diss}}^R}$ under P_F. This gives Jarzynski's (2006) criterion $M \gtrsim e^{\beta W_{\mathrm{diss}}^R}$, which is $e^{\beta W_{\mathrm{diss}}}$ for Gaussian work. Below it $\widehat{\Delta F}_J$ is biased towards $\langle W\rangle$. A stricter criterion follows from the relative variance of the weights $e^{-\beta W}$, which equals $e^{2\beta W_{\mathrm{diss}}} - 1$ for Gaussian work: the statistical error drops below $k_{\mathrm{B}}T$ only when $M \gtrsim e^{2\beta W_{\mathrm{diss}}}$. We report the effective sample size $M_{\mathrm{eff}} = (\sum w_m)^2/\sum w_m^2$ with $w_m = e^{-\beta W_m}$.

4 Exact free-energy references

The kinetic energy cancels. In bond variables $\boldsymbol{u}_i = \boldsymbol{r}_i - \boldsymbol{r}_{i-1}$ (unit Jacobian, $\boldsymbol{r}_N = \sum \boldsymbol{u}_i$) the bonds are independent, and

$$Z(\boldsymbol{\lambda}) \propto \big(g^{*N} * w\big)(\boldsymbol{\lambda}), \qquad g(\boldsymbol{u}) = e^{-\beta k(|\boldsymbol{u}|-b)^2/2}, \quad w(\boldsymbol{s}) = e^{-\beta k_t|\boldsymbol{s}|^2/2}, \tag{8}$$

i.e. the trap acts as an $(N{+}1)$-th bond with zero rest length.

Gaussian chain. Variances add (springs in series):

$$\Delta F = \tfrac{1}{2}k_{\mathrm{eff}}\big(x_1^2 - x_0^2\big), \qquad 1/k_{\mathrm{eff}} = N/k + 1/k_t, \tag{9}$$

which is independent of T. For $N = 10$: $\Delta F = 10.125$.

Stiff-bond chain. Keeping only the second moment of each bond gives the *entropic-spring* approximation $\Delta F \approx k_{\mathrm{B}}T(x_1^2 - x_0^2)/(N\langle u^2\rangle + 2k_{\mathrm{B}}T/k_t)$, which is proportional to T. The exact value comes from iterating the radial 2D convolution in log space,

$$\ln h_{n+1}(R) = \ln\int r\,\mathrm{d}r\,\mathrm{d}\theta\; g(r)\,h_n\big(\sqrt{R^2 + r^2 - 2Rr\cos\theta}\big), \tag{10}$$

with log-sum-exp quadrature ($\Delta R = 0.01$, 161 radial and 256 angular nodes). The Fourier (Hankel) route $\int \hat{g}(q)^N \hat{w}(q) J_0(q\lambda) q \, \mathrm{d}q$ is mathematically equivalent but cancels catastrophically when $Z(\boldsymbol{\lambda}_1)/Z(\boldsymbol{\lambda}_0) \lesssim 10^{-4}$; it even returned negative values at $N = 40$. The log-space method reproduces Eq. (9) to 10^{-4} (at $N = 40$: 64.796 vs. 64.800) and agrees with the Hankel route to $< 10^{-3}$ wherever that route is reliable (tested for $N = 3, 12$).

N, $k_\mathrm{B}T$	exact ΔF	entropic spring	error	$\Delta F/k_\mathrm{B}T$
5, 1	1.985	2.103	+6%	1.98
10, 1	4.420	4.286	−3%	4.42
20, 1	9.491	8.654	−9%	9.49
40, 1	19.694	17.391	−12%	19.69
10, 0.25	1.156	1.111	−4%	4.62
10, 0.5	2.276	2.195	−4%	4.55
10, 2	8.365	8.182	−2%	4.18

Table 1: Exact ΔF for the stiff-bond chain ($b = 1$, protocol $0.2L \to 0.7L$).

What the table shows. (i) $\Delta F/k_\mathrm{B}T$ is almost independent of T: elasticity is entropic. The small drift ($4.62 \to 4.18$) comes from bond stretching, which makes $\langle u^2 \rangle$ grow from 1.008 to 1.06 as T rises. (ii) At fixed relative extension ΔF is nearly *extensive*: 0.40, 0.44, 0.47, 0.49 $k_\mathrm{B}T$ per bead for $N = 5, 10, 20, 40$. (iii) The entropic spring underestimates ΔF for long chains (finite extensibility at 70% stretch). It *overestimates* it for $N = 5$, because short 2D chains are strongly non-Gaussian. A 3-bond chain even has its end-to-end density peaked near $R \approx b$ rather than at $R = 0$. The approximation's error decays like $1/N$ at small extension (unit test).

5 Numerical scheme

Integrator (BAOAB). The Langevin generator is split into kick (B), drift (A) and Ornstein–Uhlenbeck (O) parts, each solved exactly and applied in the order B–A–O–A–B:

$$\mathrm{B} : \boldsymbol{p} \leftarrow \boldsymbol{p} + \tfrac{\Delta t}{2} \boldsymbol{F}(\boldsymbol{r}; \boldsymbol{\lambda}), \qquad \mathrm{A} : \ \boldsymbol{r} \leftarrow \boldsymbol{r} + \tfrac{\Delta t}{2m} \boldsymbol{p},$$
$$\mathrm{O} : \boldsymbol{p} \leftarrow e^{-\gamma \Delta t} \boldsymbol{p} + \sqrt{(1 - e^{-2\gamma \Delta t}) m k_\mathrm{B} T} \, \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(0, \mathbb{1}).$$

The O step satisfies fluctuation–dissipation exactly for any Δt. BAOAB has the smallest configurational bias among standard splittings (Leimkuhler & Matthews 2013) and needs one force evaluation per step. The fastest mode is the bond stretch, $\omega_\mathrm{max} \approx 2\sqrt{k/m} = 20$; with $\Delta t = 0.01$, $\omega_\mathrm{max} \Delta t = 0.2$.

Work bookkeeping. At every step: (1) $W \mathrel{+}= \frac{k_t}{2}[(x_N - \lambda_{k+1})^2 - (x_N - \lambda_k)^2]$ at the current x_N; (2) one BAOAB step at $\boldsymbol{\lambda}_{k+1}$. This is literally the switch/relax scheme of the proof. A unit test checks $\Delta H = W + Q$ per trajectory to 10^{-9}.

Exact initial ensemble (tilted rejection sampling). Every pull must start from the equilibrium ensemble at its initial trap position. Instead of equilibrating each chain dynamically, we draw exact equilibrium configurations (theory document, Sec. 10.2):

1. *Target.* In bond variables the Boltzmann weight is $\prod_i g(\boldsymbol{u}_i) \times w(\boldsymbol{\lambda} - \boldsymbol{R})$, $\boldsymbol{R} = \sum_i \boldsymbol{u}_i$. The bonds would be independent if it were not for the trap factor w, which only depends on the end point.

2. *Plain rejection.* Draw a free chain (independent bonds) and keep it with probability $w(\boldsymbol{\lambda}-\boldsymbol{R}) \leq 1$, the trap's Boltzmann factor. Kept chains are exact samples, but a free chain rarely ends where a stretched one should. For the reverse pulls at $N = 40$ ($\lambda = 28$, while free chains end at $|\boldsymbol{R}| \sim 6$) only $\sim 10^{-9}$ of the proposals are kept.
3. *Tilt.* Draw instead chains pulled by a constant force f along x, density $\prod_i g(\boldsymbol{u}_i)e^{\beta f u_{i,x}}$. The bonds remain independent and easy to draw (length by inverse CDF of $r\,g(r)I_0(\beta f r)$, angle from a von Mises law), and the proposed chains are already stretched.
4. *Correction.* The ratio target/proposal is $w(\boldsymbol{\lambda} - \boldsymbol{R})e^{-\beta f R_x} \propto e^{-\beta k_t |\boldsymbol{R}-\boldsymbol{\lambda}'|^2/2} \leq 1$ with $\boldsymbol{\lambda}' = \boldsymbol{\lambda} - (f/k_t)\hat{\boldsymbol{x}}$, so the chain is kept with that probability. This is exact for any f; f only changes the efficiency.
5. *Choice of f.* f is set to the chain tension at that extension (proposal mean end $= \boldsymbol{\lambda}'$). The acceptance rate rises to about 1% at $N = 40$ and 2–3% at $N = 10$.

There is thus no equilibration run and no equilibration error. The forward runs of the first version used $f = 0$, which samples the same distribution exactly but more slowly.

Implementation and cost. The code is in Python. Sampling is done in NumPy; the integrator runs in PyTorch (float64) on the GPU (RTX 2060), vectorised over the M independent replicas, with a cost of about 2–15 ms per step for $M = 10^5$. The largest run ($N = 40$, 3.4×10^4 steps, 10^5 chains) took 16 minutes including sampling. All runs are cached in `data/runs/`.

Estimators. For each parameter set we report $\langle W\rangle$, σ_W, $\widehat{\Delta F}_J$, the second-cumulant estimate $\langle W\rangle - \beta\sigma_W^2/2$ and M_{eff}. Error bars are bootstraps over pulls (but see Sec. 6.6). Convergence is shown by splitting the 10^5 pulls into disjoint blocks of n and plotting the mean and spread of $\widehat{\Delta F}_J$ over blocks.

Unit tests (pytest, 20 tests). The suite checks:

- forces equal $-\nabla H$ (autograd), for both chains;
- the sampler reproduces the exact Gaussian-chain mean and covariance (also at full stretch), the free-bond $\langle u^2\rangle$, and tilted and untilted samplers agree;
- exact Crooks relations for the Gaussian chain (equal forward/reverse variances, $\langle W\rangle_F - \Delta F = \langle W\rangle_R + \Delta F = \beta\sigma^2/2$), and BAR/reverse estimators on synthetic Crooks pairs;
- the sampled mean trap force equals $\partial_\lambda F$ from the exact reference;
- the log-space reference agrees with the closed form and with the Hankel route;
- BAOAB at a fixed trap position preserves the exact equilibrium;
- the first law holds per trajectory;
- the simulated work moments match the exact linear theory;
- Jarzynski holds exactly (to 10^{-3}) for the Gaussian chain at three pulling times;
- the estimators behave correctly on synthetic data.

Exact work statistics of the discrete algorithm ($b = 0$). With linear forces, $(\boldsymbol{q}, \boldsymbol{p}, W)$ along x is a linear-Gaussian Markov chain: each sub-step (switch, B, A, O) is an affine map plus Gaussian noise. Propagating the mean vector and covariance matrix (21×21 for $N = 10$) through all steps gives the *exact* $\langle W\rangle$ and $\operatorname{Var} W$ of the simulated algorithm, including all Δt effects.

5.1 Time-step error

For the Gaussian chain the exact linear propagation gives the residual $\langle W\rangle - \beta\sigma_W^2/2 - \Delta F$ directly. It is -8.0×10^{-4}, -3.0×10^{-3} and $-1.1\times10^{-2}\,k_{\mathrm{B}}T$ at $\Delta t = 0.01, 0.02, 0.04$ ($\tau = 0.2$), a clean $O(\Delta t^2)$ law. The residual is not zero, although BAOAB samples harmonic configurations exactly, because the scheme's *momentum* distribution is slightly distorted while the proof requires the full phase-space measure to be preserved. For the stiff-bond chain ($N = 10$, $\tau = 4\tau_1$, 10^5 pulls per run), halving Δt from 0.01 to 0.005 changed the three-seed mean of $\beta\widehat{\Delta F}_J$ by -0.07 ($k_{\mathrm{B}}T = 2$) and -0.05 ($k_{\mathrm{B}}T = 1$). This is comparable to the seed-to-seed scatter of single estimates (0.05–0.07), and $\langle W\rangle$ changed by less than 0.1%. No time-step effect is detectable. The time step is therefore not a limitation at the accuracy reached here.

6 Results

6.1 Validation on the Gaussian chain (F1)

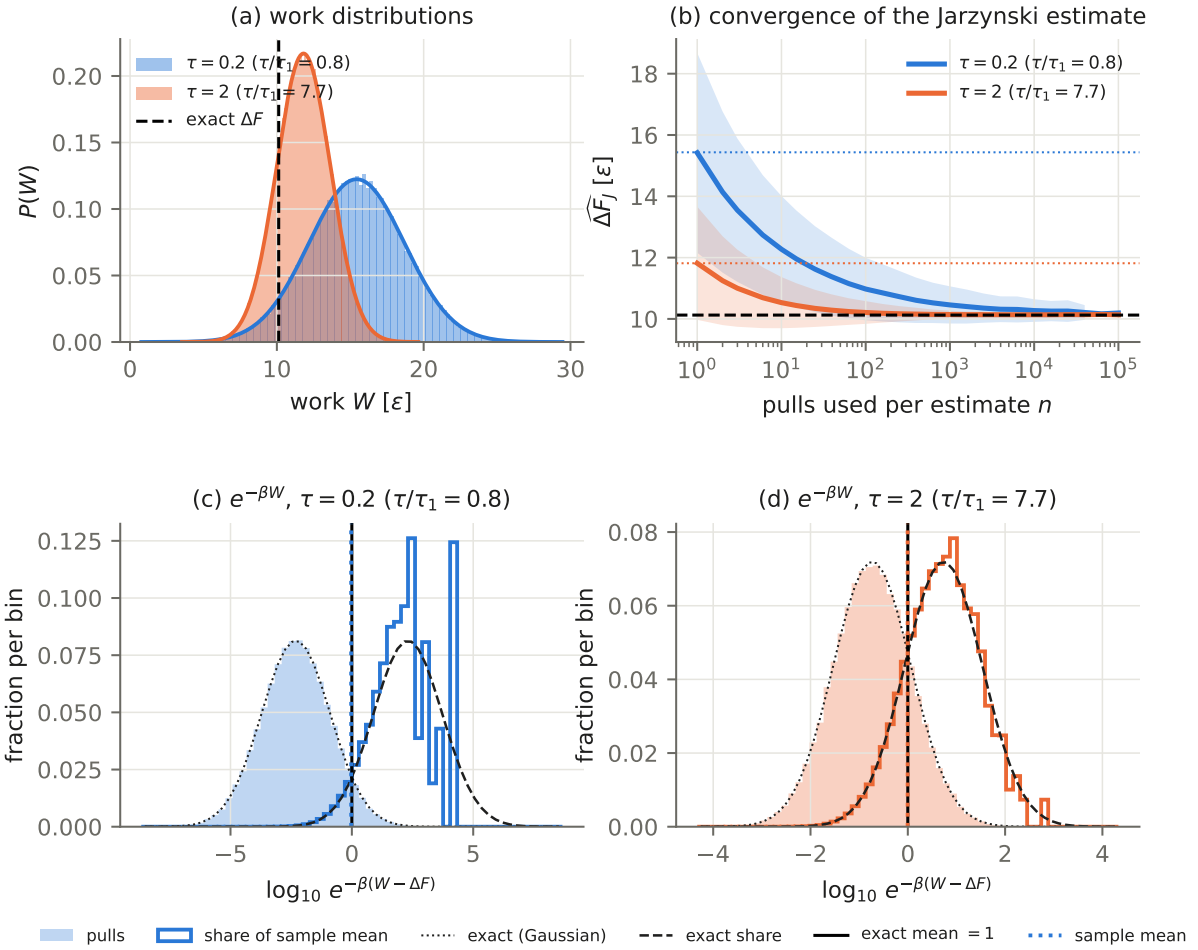


Figure 2: Gaussian chain, $N = 10$, $k_{\mathrm{B}}T = 1$, 10^5 pulls per speed. (a) Simulated work histograms; lines are the exact Gaussian distribution with moments from the linear propagation; dashed line: exact $\Delta F = 10.125$. (b) Jarzynski estimate using n pulls: mean (line) and ±1 s.d. (band) over disjoint blocks; dotted lines: $\langle W\rangle$. (c, d) Distribution of the Jarzynski weight $x = e^{-\beta W}/e^{-\beta\Delta F}$ (log scale), whose exact mean is 1. Filled: fraction of pulls; solid step: share of the sample mean contributed by each bin ($x\times$count$/\sum x$); dashed: exact share (Gaussian theory); dotted: exact distribution of pulls; vertical lines: exact mean 1 (black) and sample mean (coloured, dotted).

τ (τ/τ_1)	exact ΔF	$\langle W\rangle$ sim / exact	σ_W^2 sim / exact	$\beta W_{\rm diss}$	$\widehat{\Delta F}_J$	$M_{\rm eff}$
0.2 (0.8)	10.125	15.420 / 15.435	10.64 / 10.62	5.30	10.19 ± 0.14	51
2 (7.7)	10.125	11.816 / 11.816	3.41 / 3.38	1.69	10.134 ± 0.015	4348

Reading. Simulation and exact theory agree within statistical error, both for the moments and for the whole distribution (Fig. 2a). $\widehat{\Delta F}_J$ recovers $\Delta F = 10.125$ for both speeds, even though the average work overshoots it by $5.3\,k_{\rm B}T$ for the fast pull. Panel (b) shows the characteristic finite-sample behaviour: with a handful of pulls the estimate sits near $\langle W\rangle$, and it converges only once n exceeds a few hundred to a few thousand pulls. The faster pull needs about 100× more pulls, consistent with the $e^{\beta W_{\rm diss}}$ to $e^{2\beta W_{\rm diss}}$ scaling.

Where the exponential average comes from (panels c, d). The weight $x = e^{-\beta(W-\Delta F)}$ has mean exactly 1, but its distribution is extremely skewed. Most pulls have $x \ll 1$ (they dissipate), and the mean is carried by a few pulls with $x \gg 1$. The contribution to the mean ("share") is shifted relative to where the pulls are, by $\beta^2\sigma_W^2/\ln 10$ decades for Gaussian work. For the slow pull (d) the shift is 1.5 decades, the two histograms overlap strongly, and the sampled share follows the exact one. For the fast pull (c) the shift is 4.6 decades. The exact share peaks at $x \approx 10^{2.3}$, which only 0.06% of pulls reach (about 60 of 10^5), and extends to $x \sim 10^7$. The sampled share is a handful of spikes that stops near $x \approx 10^4$, so the sample mean rests on a few dozen pulls ($M_{\rm eff} = 51$). That it still lands within $0.07\,k_{\rm B}T$ of the truth is helped by the Gaussian tail decaying fast.

6.2 Effect of the pulling speed (F2)

τ/τ_1	exact ΔF	$\langle W\rangle$	σ_W	skew	$\beta W_{\rm diss}$	affine est.	$\widehat{\Delta F}_J$	$\langle W\rangle - \frac{\beta}{2}\sigma_W^2$	$M_{\rm eff}$
1	4.420	22.71	7.52	0.21	18.3	16.4	4.99 ± 0.48	−5.54	4.9
3	4.420	10.58	3.86	0.16	6.16	5.48	4.55 ± 0.10	3.12	103
30	4.420	5.03	1.12	0.05	0.61	0.55	4.415 ± 0.005	4.404	30784

Dissipation law. In a slow pull the chain stretches affinely: bead i moves with velocity $\approx v\,i/N$, $v = (x_1 - x_0)/\tau$. The friction power is then $\sum_i \zeta(vi/N)^2 \approx \zeta N v^2/3$. With $x_1 - x_0 = N/2$ and Eq. (3),

$$\beta W_{\rm diss} \simeq \frac{\zeta N (x_1 - x_0)^2}{3 k_{\rm B} T \tau} = \frac{\pi^2}{6}\,\frac{N}{\tau/\tau_1}. \tag{11}$$

The simulation exceeds this estimate by 11–12% at every speed, so the scaling $W_{\rm diss} \propto 1/\tau$ holds down to $\tau = \tau_1$. The near-equilibrium relation $\sigma_W^2 = 2k_{\rm B}TW_{\rm diss}$ holds to 3% at $\tau/\tau_1 = 30$, but fails by ×1.2 and ×1.5 for the faster pulls: the work distribution becomes broader than linear response predicts and positively skewed. At $\tau = \tau_1$ the second-cumulant estimate is wrong by $10\,k_{\rm B}T$, while $\widehat{\Delta F}_J$ is still within $0.6\,k_{\rm B}T$ because a handful of pulls ($M_{\rm eff} \approx 5$) happened to land near ΔF. Its block-averaged curve (Fig. 3b) has not converged at $n = 10^5$, as expected for $e^{18} \approx 10^8$.

A rare trajectory can push the estimate below ΔF. In a pilot run with 10^4 pulls at $\tau/\tau_1 = 3$, $\widehat{\Delta F}_J = 2.62$, which is *below* $\Delta F = 4.42$, with $M_{\rm eff} = 1.9$. The estimate was dominated by one pull with $W = -6.3$, a 4.4σ event. We traced that trajectory. The end bead started 0.76 ahead of the trap centre (a 2.4σ thermal fluctuation) and the chain stayed ahead of the trap for the whole pull: the bath did the stretching and the trap was *pushed*. Such pulls are rarer than a Gaussian model suggests: in the 10^5-pull runs at $\tau/\tau_1 = 1$ and 3, only 4 pulls fall below $\langle W\rangle - 3.5\sigma_W$, against 23 for a Gaussian. The right tail is heavier (65 and 62 pulls above $+3.5\sigma_W$), hence the positive skew. A positive third cumulant is also why the second-cumulant estimate falls *below* ΔF in the table. The usual statement that the Jarzynski estimate is "biased upwards" is true on average, but a single under-sampled estimate can fail in either direction. With 10^5 pulls the estimate is 4.55 ± 0.10.

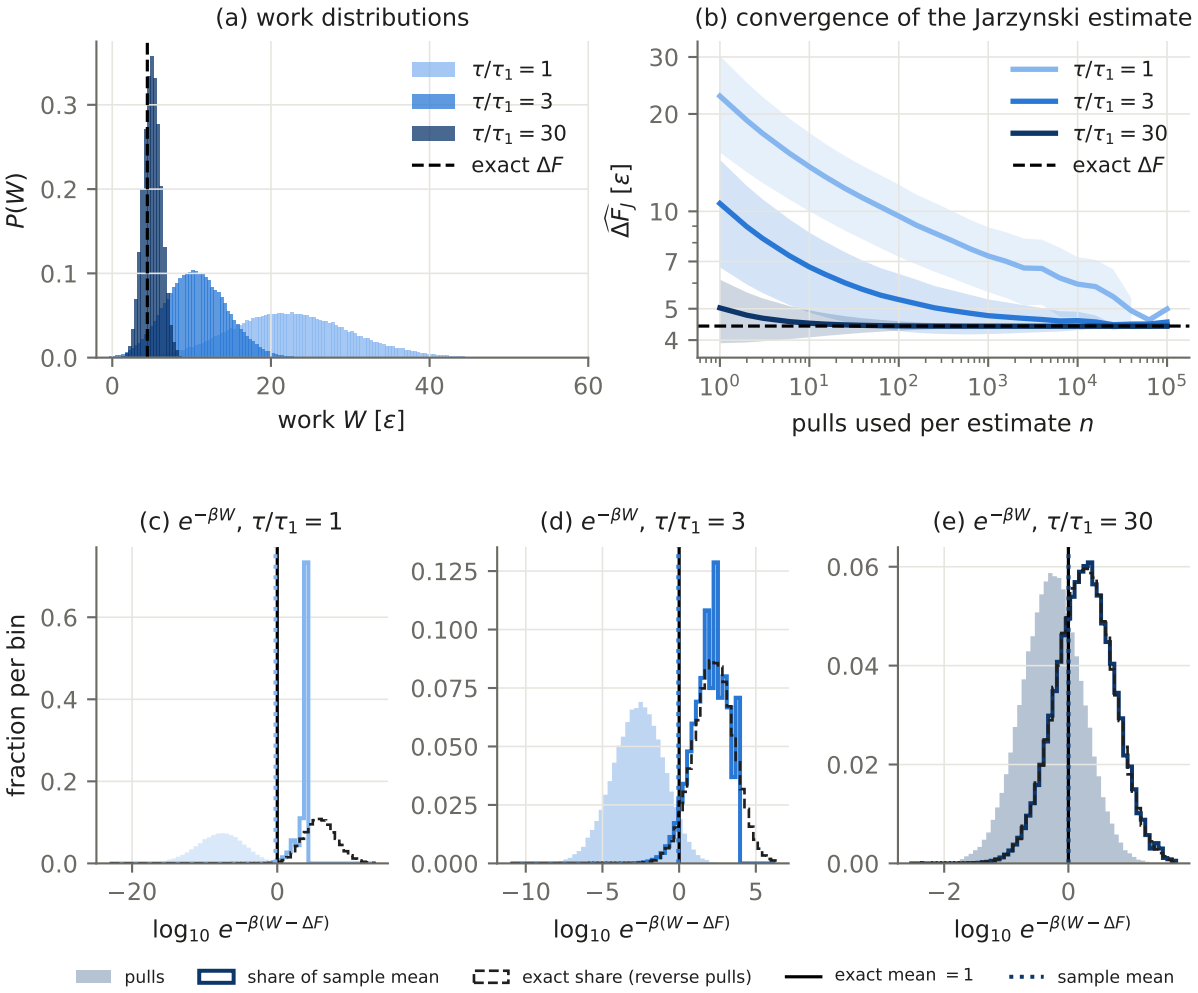


Figure 3: Stiff-bond chain, $N = 10$, $k_\mathrm{B}T = 1$, $\tau_1 = 5.46$, 10^5 pulls per speed. (a) Work distributions; dashed: exact $\Delta F = 4.420$. (b) Block-averaged Jarzynski estimate using n pulls (line: mean over blocks, band: ± 1 s.d.). (c–e) Distribution of $x = e^{-\beta(W-\Delta F)}$ as in Fig. 2. Here the exact share of the mean (dashed) comes from 10^5 *reverse* pulls through Crooks' identity (7).

Seeing the mean of $e^{-\beta W}$ (panels c–e). Thanks to Crooks, the reverse pulls give the *exact* shape of the share of $\langle e^{-\beta W} \rangle$ without any forward rare-event sampling.

- $\tau/\tau_1 = 30$: half of the true mean comes from $x > 10^{0.27}$, a region reached by 14% of forward pulls. The sampled share matches the exact one, and no single pull carries more than 0.1% of the mean.
- $\tau/\tau_1 = 3$: half of the mean comes from $x > 10^{2.3}$, reached by 0.07% of pulls (≈ 70). The sampled share follows the exact one only on its lower half, and the largest pull carries 7% of the mean.
- $\tau/\tau_1 = 1$: half of the true mean comes from $x > 10^{5.9}$, but the largest weight among all 10^5 pulls is $10^{4.3}$. *No* pull reaches the region that dominates the average. The sample mean is carried by three pulls (33%, 27%, 14%) at the edge of the sampled range, and it equals $e^{-\beta(4.99-4.42)} = 0.57$ of the true mean, consistent with the missing upper half. The estimate still looks deceptively good ($+0.57\,k_\mathrm{B}T$) because the logarithm compresses errors: missing half of $\langle e^{-\beta W} \rangle$ costs only $k_\mathrm{B}T \ln 2 \approx 0.7\,k_\mathrm{B}T$. Nothing in the forward sample itself reveals that half of the average was never seen; the reverse pulls do.

6.3 Temperature (F3)

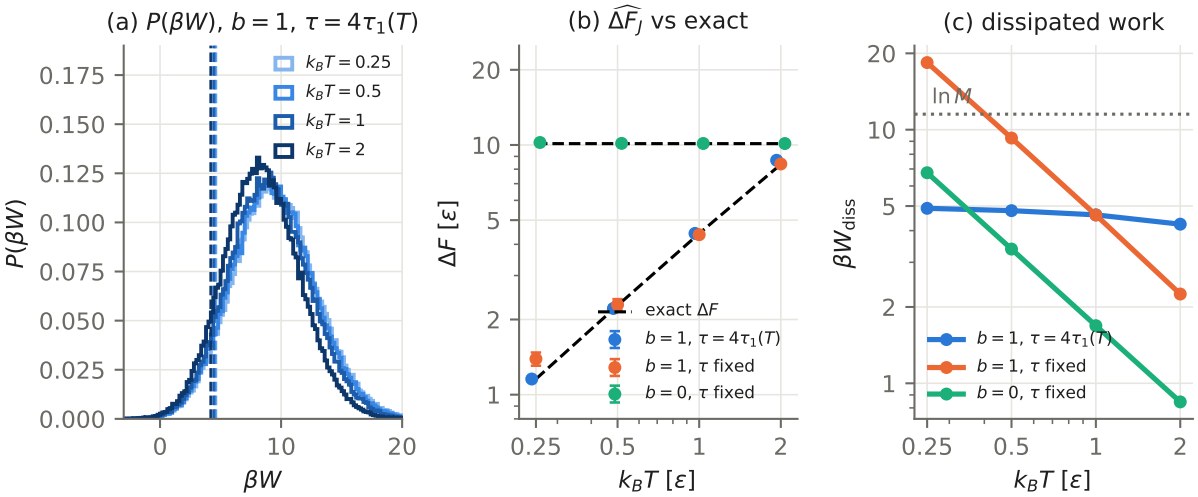


Figure 4: Temperature sweep, $N = 10$, 10^5 pulls per point. (a) Stiff-bond chain at fixed reduced speed $\tau = 4\tau_1(T)$: distributions of βW (dashed: $\beta\Delta F$). (b) Jarzynski estimates (bootstrap error bars) against the exact ΔF (dashed), for the stiff-bond chain at fixed τ/τ_1 and at fixed $\tau = 21.9 = 4\tau_1(k_BT{=}1)$, and for the Gaussian chain at fixed $\tau = 2$. (c) βW_{diss} for the same three series; dotted: $\ln M$.

	$b = 1$, $\tau = 4\tau_1(T)$				$b = 1$, $\tau = 21.9$ fixed			
k_BT	ΔF	$\widehat{\Delta F}_J$	βW_{diss}	M_{eff}	$\widehat{\Delta F}_J$	βW_{diss}	M_{eff}	$\beta(\widehat{\Delta F}_J - \Delta F)$
0.25	1.156	1.155 ± 0.014	4.90	307	1.39 ± 0.08	18.4	9	+0.94
0.5	2.276	2.214 ± 0.055	4.80	81	2.29 ± 0.12	9.27	17	+0.03
1	4.420	4.425 ± 0.055	4.61	331	4.37 ± 0.06	4.61	267	−0.05
2	8.365	8.683 ± 0.082	4.24	648	8.394 ± 0.033	2.25	3336	+0.01

Entropic vs. energetic elasticity. For the stiff-bond chain ΔF is proportional to T: a hotter chain is *harder* to stretch, as in rubber. For the Gaussian chain $\Delta F = 10.125$ at every temperature (Fig. 4b), and its measured $\langle W\rangle = 11.815 \pm 0.002$ is also temperature-independent. In a linear system the mean response is deterministic, and the bath only adds variance, $\sigma_W^2 \propto T$; hence $\beta W_{\mathrm{diss}} \propto 1/T$ exactly (green line in Fig. 4c, matching the linear theory to 0.3%).

Collapse at fixed reduced speed. When the pulling time is scaled with the relaxation time, $\tau = 4\tau_1(T)$, the four distributions of βW fall on top of each other (Fig. 4a) and $\beta W_{\mathrm{diss}} \approx 4.2$–$4.9$ is nearly constant. This follows from Eq. (11): $W_{\mathrm{diss}} \propto \zeta/\tau \propto T$ and $\Delta F \propto T$, so temperature is just a change of energy unit for the entropic chain. The residual drift comes from the slight bond stretching at high T (Table 1).

At fixed pulling speed, cold baths are harder. At fixed τ, W_{diss} is set by friction and speed and is nearly T-independent, so $\beta W_{\mathrm{diss}} \propto 1/T$ (orange line: $18.4, 9.3, 4.6, 2.2$, an exact factor of 2 per halving of T). At $k_BT = 0.25$, βW_{diss} exceeds $\ln M = 11.5$ and Jarzynski fails: $\widehat{\Delta F}_J$ is too high by $0.94\,k_BT$ (20%). The same pulling experiment at $k_BT \geq 0.5$ recovers ΔF within $0.05\,k_BT$.

6.4 Number of beads (F4)

N	τ_1	ΔF	$\langle W\rangle$	σ_W	βW_{diss}	affine (11)	$\widehat{\Delta F}_J$ (10^5 pulls)	$\beta(\widehat{\Delta F}_J - \Delta F)$	M_{eff}
5	1.45	1.985	5.04	2.64	3.05	2.06	2.018 ± 0.025	+0.03	1239
10	5.46	4.420	9.03	3.28	4.61	4.11	4.429 ± 0.059	+0.01	252
20	21.3	9.491	17.78	4.43	8.29	8.22	9.87 ± 0.11	+0.38	84
40	84.3	19.694	35.65	6.17	15.96	16.45	21.16 ± 0.55	+1.47	3.6

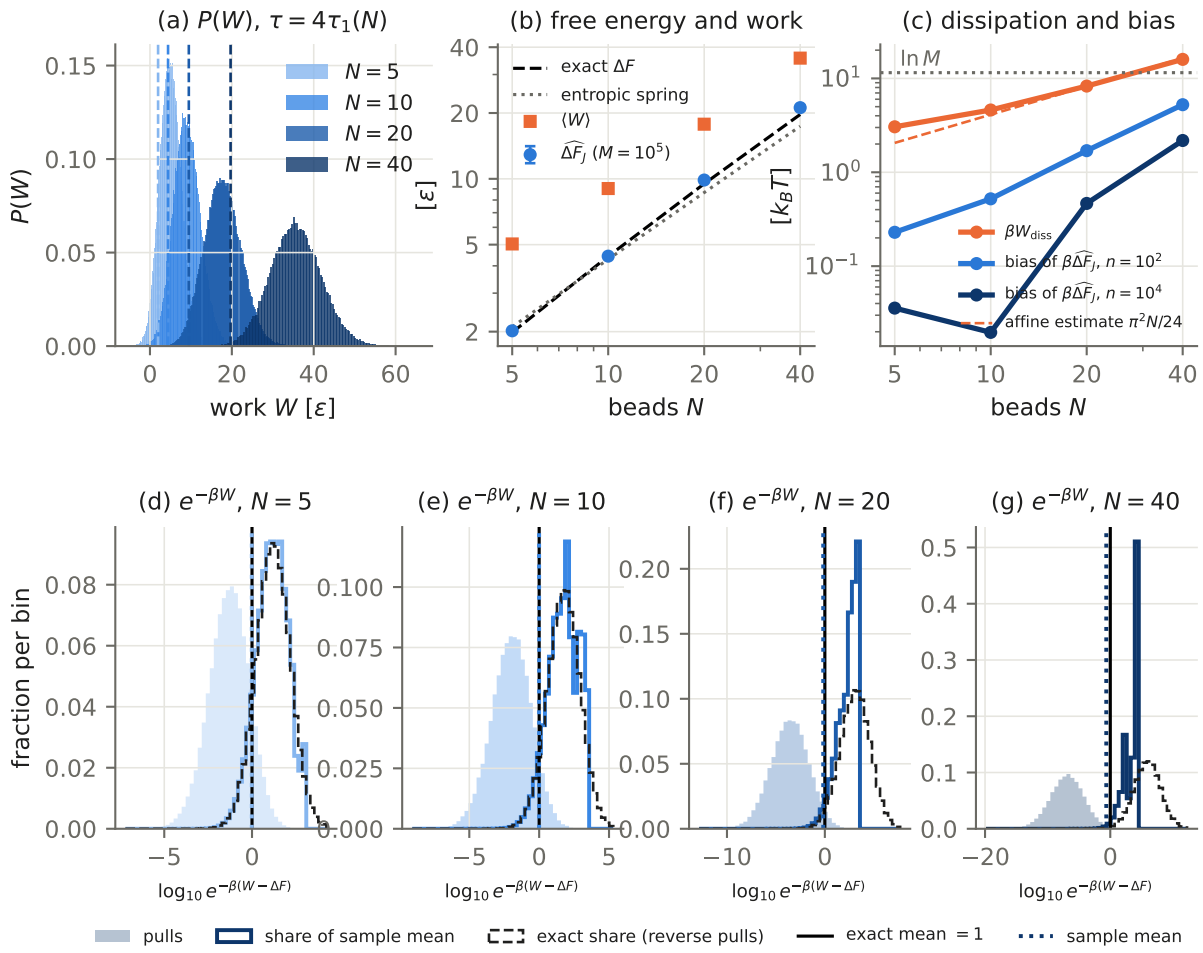


Figure 5: Stiff-bond chain, $k_{\mathrm{B}}T = 1$, fixed relative extension $0.2L \to 0.7L$, fixed reduced speed $\tau = 4\tau_1(N)$, 10^5 pulls per point. (a) Work distributions (dashed: exact ΔF). (b) Exact ΔF, entropic-spring approximation, $\langle W \rangle$ and $\widehat{\Delta F}_J$ using all 10^5 pulls. (c) βW_{diss} and the bias of $\beta\widehat{\Delta F}_J$ when only n pulls are available (mean over disjoint blocks). (d–g) Distribution of $x = e^{-\beta(W-\Delta F)}$ for each N, as in Fig. 3; the exact share of the mean (dashed) comes from reverse pulls.

Free energy and work are both extensive. At fixed relative extension, ΔF and $\langle W \rangle$ grow nearly linearly with N (Fig. 5b), while σ_W grows like $\sqrt{N}$ (successive ratios $1.24, 1.35, 1.39 \to \sqrt{2}$). The work is a sum of many weakly correlated contributions along the chain. Its *relative* fluctuations therefore shrink like $1/\sqrt{N}$, which is the approach to the macroscopic limit where $W \to \langle W \rangle$ is sharp.

Dissipation grows linearly with N at fixed reduced speed. $\beta W_{\mathrm{diss}} = 3.05, 4.61, 8.29, 15.96$ roughly doubles each time N doubles. It approaches the affine estimate $\pi^2 N/24$ (within 1% at $N = 20$, 3% at $N = 40$). For $N = 5$ the estimate is 1/3 too low, because the continuum picture of an affinely stretching chain is poor for five bonds.

Consequence: exponential cost in system size. The number of pulls needed, $e^{\beta W_{\mathrm{diss}}}$, is ≈ 20, 10^2, 4×10^3 and 9×10^6 for $N = 5, 10, 20, 40$. With $M = 10^5$ the estimate is accurate up to $N = 10$. It is biased by $+0.38\, k_{\mathrm{B}}T$ at $N = 20$, where $e^{\beta W_{\mathrm{diss}}} < M < e^{2\beta W_{\mathrm{diss}}}$, and fails at $N = 40$ ($+1.47\, k_{\mathrm{B}}T$, $M_{\mathrm{eff}} = 3.6$). Panel (c) shows the bias at fixed budget: it grows steeply with N for every n, and for $N = 40$ it is still $2.2\, k_{\mathrm{B}}T$ with 10^4 pulls. The bootstrap error bars miss the exact value by 3.4σ ($N = 20$) and 2.7σ ($N = 40$), the same over-confidence as in Sec. 6.6. Keeping βW_{diss} fixed while N grows would require $\tau \propto N\tau_1 \propto N^3$. This is why Jarzynski-based free energies are a tool for single molecules and small systems.

The breakdown seen in $e^{-\beta W}$ (panels d–g). As N grows, the pull histogram moves left (more

dissipation) and the exact share of the mean moves right, so the two separate. At $N = 5$ and 10 they overlap and the sampled share follows the exact one. At $N = 20$ the sampled share is cut off at the largest observed weight ($10^{3.5}$), and 40% of the exact share lies beyond it. At $N = 40$ one pull carries 51% of the sample mean and the next 13%, and 81% of the exact share lies beyond the largest forward weight ($10^{4.1}$). Most of the average was never sampled.

6.5 Crooks fluctuation theorem and two-sided estimation (F5)

For each forward run of F2 and F4 we ran 10^5 *reverse* pulls: the trap moves from $0.7L$ back to $0.2L$ in the same time τ, starting from the exact equilibrium ensemble at $0.7L$ (tilted sampler). The reverse works W_R are compared with the forward works through Eq. (7).

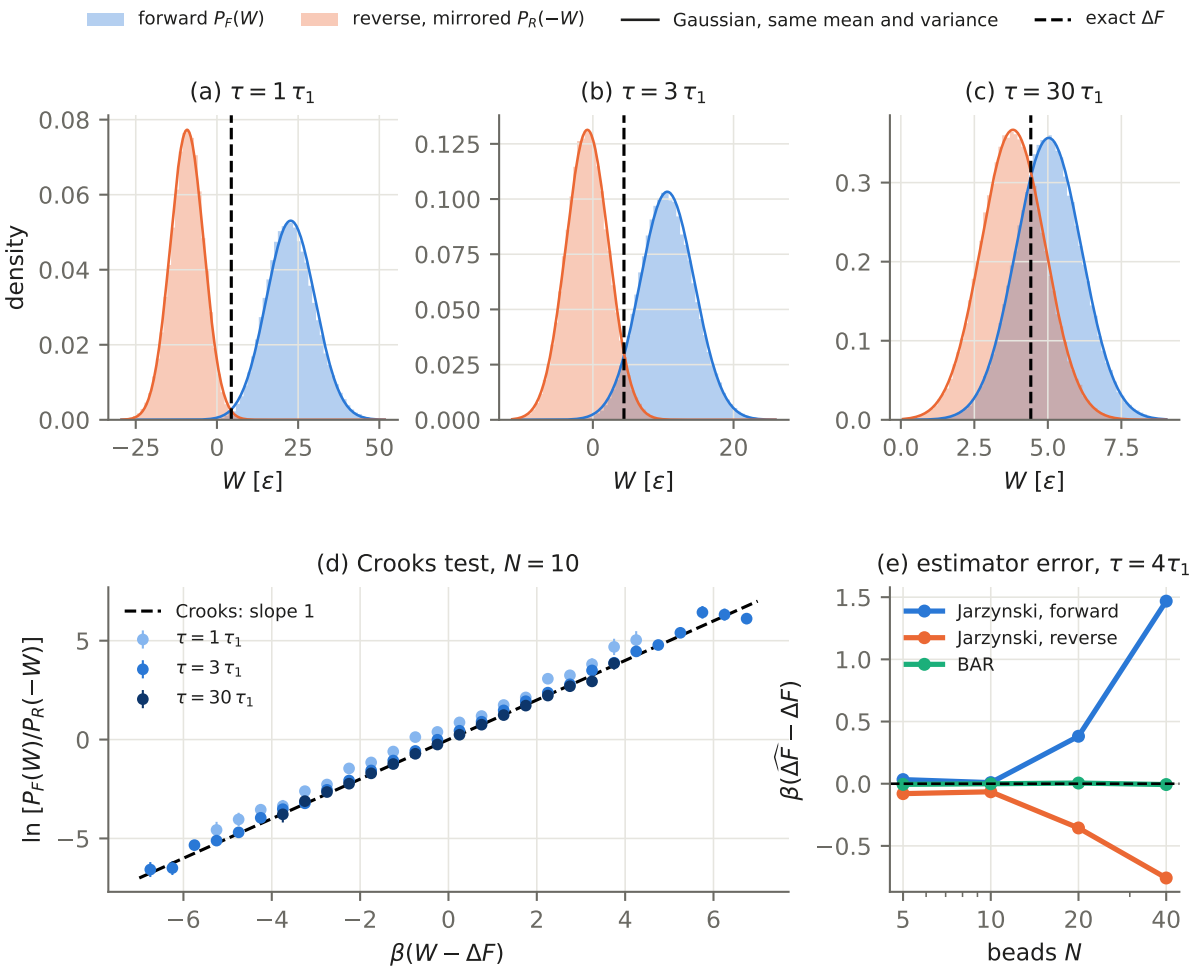


Figure 6: Crooks theorem, stiff-bond chain, $k_BT = 1$, 10^5 forward and 10^5 reverse pulls per point. (a–c) $N = 10$: forward work distribution $P_F(W)$ (blue) and mirrored reverse distribution $P_R(-W)$ (orange) for $\tau = \tau_1,\ 3\tau_1,\ 30\tau_1$. Lines are Gaussians with the same mean and variance; the dashed line is the exact $\Delta F = 4.420$, where the two distributions cross. (d) $\ln[P_F(W)/P_R(-W)]$ against $\beta(W - \Delta F)$ in the overlap region (bins with Poisson error < 0.5). Crooks predicts the dashed line of slope 1 through the origin, for every speed. (e) Error of forward Jarzynski, reverse Jarzynski and BAR against N at $\tau = 4\tau_1$.

τ/τ_1	exact ΔF	$\langle W_F \rangle$	$\langle W_R \rangle$	$\beta W^F_{\mathrm{diss}}$	$\beta W^R_{\mathrm{diss}}$	Crooks slope	$\widehat{\Delta F}_J$ fwd	$\widehat{\Delta F}_J$ rev	BAR
1	4.420	22.71	9.21	18.3	13.6	0.985	4.99	3.88	4.340 ± 0.039
3	4.420	10.58	0.79	6.16	5.21	0.994	4.55	4.40	4.414 ± 0.007
30	4.420	5.03	−3.81	0.61	0.61	0.983	4.415	4.417	4.416 ± 0.002

N	exact ΔF	$\beta W^F_{\rm diss}$	$\beta W^R_{\rm diss}$	$\beta(\widehat{\Delta F}^{\rm fwd}_J - \Delta F)$	$\beta(\widehat{\Delta F}^{\rm rev}_J - \Delta F)$	$\beta(\widehat{\Delta F}_{\rm BAR} - \Delta F)$
5	1.985	3.05	2.76	+0.033	−0.080	-0.005 ± 0.007
10	4.420	4.61	4.03	+0.009	−0.065	-0.000 ± 0.007
20	9.491	8.29	7.10	+0.382	−0.356	$+0.006 \pm 0.013$
40	19.694	15.96	13.59	+1.468	−0.758	-0.007 ± 0.039

(BAR error bars are the spread over 10 disjoint sub-samples divided by $\sqrt{10}$.)

Crooks holds. For every speed $P_F(W)$ and $P_R(-W)$ cross at ΔF (Fig. 6a–c). The log-ratio is a straight line with slope 0.983–0.994, independent of the speed (Fig. 6d), although the two distributions themselves change completely between $\tau = \tau_1$ and $30\tau_1$. The residual 1–2% is of the size expected from finite histogram bins and was not investigated further. At $\tau = \tau_1$ the points lie slightly above the line, consistent with the small negative BAR error below: the overlap region is thin there.

Shapes of the forward and reverse distributions. Both are close to Gaussian at every speed (Fig. 6a–c), with skewness at most 0.21 and excess kurtosis at most 0.10. The reverse distribution is the *narrower* one:

τ/τ_1	σ_F	σ_R	skew$_F$	skew$_R$	$\sigma_F^2/2k_{\rm B}TW^F_{\rm diss}$	$\sigma_R^2/2k_{\rm B}TW^R_{\rm diss}$
1	7.52	5.16	+0.21	+0.15	1.54	0.98
3	3.86	3.04	+0.16	−0.07	1.21	0.89
30	1.12	1.09	+0.05	−0.05	1.03	0.98

The last two columns test the linear-response relation $\sigma_W^2 = 2k_{\rm B}TW_{\rm diss}$, which holds exactly for Gaussian work near equilibrium. The reverse pull satisfies it within about 10% even at $\tau = \tau_1$, while the forward pull exceeds it by 54%. So the *forward* pull is the more strongly nonlinear process. A plausible reading, not tested further: in a fast forward pull the trap drags the end into the stiff, finite-extensibility regime, where small differences in the chain's shape produce large differences in the force and hence in the work. In a reverse pull the tension is being released and the chain moves towards the soft, nearly Gaussian regime. Because the reverse mean work is small ($\langle W_R\rangle = -\Delta F + W^R_{\rm diss}$, e.g. 0.79 at $3\tau_1$), its spread looks large *relative to its mean*, and individual reverse works change sign. That is a cancellation between released free energy and dissipation, not larger fluctuations. These shapes are measured from 10^5 pulls each and are well sampled. The sampling difficulty discussed above concerns only the far tails that dominate the exponential averages, not the bulk of the distributions.

Forward and reverse dissipation differ. The reverse pull dissipates less: $\beta W^R_{\rm diss} = 13.6$ vs. $\beta W^F_{\rm diss} = 18.3$ at $\tau = \tau_1$, and 13.6 vs. 16.0 at $N = 40$. They become equal only near equilibrium ($\tau = 30\tau_1$), where both distributions are Gaussian and Crooks forces $W^F_{\rm diss} = W^R_{\rm diss}$. A plausible reason, not tested here, is that the reverse pull starts from the stretched chain, which is stiffer (finite extensibility) and therefore relaxes faster and keeps up with the trap better. This asymmetry matters because, by Eq. (7), forward Jarzynski needs $M \gtrsim e^{\beta W^R_{\rm diss}}$ pulls (8×10^5 at $N = 40$, more than we have), and reverse Jarzynski needs $e^{\beta W^F_{\rm diss}}$. The two one-sided estimators are biased in opposite directions, as expected.

BAR fixes the size problem. Combining both directions, BAR is within $0.01\,k_{\rm B}T$ of the exact ΔF for every N, including $N = 40$, where forward and reverse Jarzynski are off by +1.47 and $-0.76\,k_{\rm B}T$ with the same 10^5 pulls each. BAR needs only the *overlap* of $P_F(W)$ and $P_R(-W)$ around ΔF, not the far tail of either. At $N = 40$ that overlap contains 0.4% of the forward pulls (those with $W < \Delta F$) and 0.3% of the reverse pulls, a few hundred each, which is enough. At $\tau = \tau_1$ ($N = 10$) the overlap is thinner and BAR is 2σ low ($-0.08\,k_{\rm B}T$), still far better than either one-sided estimate (+0.57, −0.54). At $\tau = 30\tau_1$ all three estimators agree on 4.416, $0.004\,k_{\rm B}T$ (0.1%) below the reference, which is at the level of the time-step error.

6.6 Bootstrap error bars are over-confident

In two cases the bootstrap error bar on $\widehat{\Delta F}_J$ does not cover the exact value: the Gaussian chain at $k_\mathrm{B}T = 0.25$ ($\beta W_\mathrm{diss} = 6.8$; $+0.42\,k_\mathrm{B}T$, bootstrap $0.10\,k_\mathrm{B}T$) and the stiff-bond chain at $k_\mathrm{B}T = 2$, $\tau = 4\tau_1$ ($+0.16\,k_\mathrm{B}T$, bootstrap $0.04\,k_\mathrm{B}T$). For the Gaussian case the true sampling distribution of $\widehat{\Delta F}_J$ is computable: drawing 300 samples of 10^5 Gaussian works with the exact moments gives a mean bias of $+0.10\,k_\mathrm{B}T$ and a spread of $0.28\,k_\mathrm{B}T$. The observed $+0.42$ is therefore a 1.1σ event, but the bootstrap reports a spread $3\times$ smaller than the truth. The bootstrap resamples only the pulls that were observed, so it cannot account for the rare low-W pulls that were never drawn. For the stiff-bond chain at $k_\mathrm{B}T = 2$, three further independent runs gave $+0.03$, -0.06 and $-0.00\,k_\mathrm{B}T$. The first value $+0.16\,k_\mathrm{B}T$ was therefore a fluctuation: across the four runs the estimate scatters by $0.09\,k_\mathrm{B}T$, twice the bootstrap error (0.04–$0.06\,k_\mathrm{B}T$). *Practical rule:* the error bar of a single Jarzynski estimate should come from independent repetitions (or from the block curve), not from a bootstrap of one sample, whenever M_eff is below a few thousand.

7 Discussion

- **The equality is exact; the estimator is the bottleneck.** In every case where $M \gg e^{\beta W_\mathrm{diss}}$, $\widehat{\Delta F}_J$ agrees with the exact ΔF, including pulls where the average work exceeds ΔF by $6\,k_\mathrm{B}T$. In practice the whole difficulty is sampling the left tail of $P(W)$.
- **One dimensionless number controls everything:** $\beta W_\mathrm{diss} \simeq (\pi^2/6)\,N/(\tau/\tau_1)$. Temperature enters only through $\tau_1 \propto 1/T$. Changing T at fixed τ means changing the reduced speed, which is why cold experiments are harder.
- **Size.** At a fixed reduced speed the dissipation is extensive in N, so the number of pulls needed grows exponentially with system size. To keep βW_diss fixed, τ must grow like $N\tau_1 \propto N^3$.
- **Cheap diagnostics are misleading** in the dangerous regime: the second-cumulant formula fails as soon as $P(W)$ is skewed, and the bootstrap underestimates the error by a factor ~ 3. M_eff and the block-convergence curve are the useful diagnostics: a curve still drifting at the largest n is unreliable.
- **Use both directions when possible.** The exponential average is dominated by pulls typical of the reverse process, so one-sided estimates silently miss part of it. When reverse pulls are available, BAR needs only the overlap region and gave ΔF to $0.01\,k_\mathrm{B}T$ at $N = 40$ with the same data that left one-sided Jarzynski $1.5\,k_\mathrm{B}T$ off.

Possible extensions (not done, suggested).

- *Lower dissipation by design*: optimise the protocol $\lambda(t)$ (not just its duration) to minimise $\langle W \rangle$ or the variance of the estimator. In linear response the optimum moves at constant "thermodynamic speed". Feedback control (a protocol that reacts to the measured chain) can even lower $\langle W \rangle$ below ΔF, at the cost of information (Sagawa–Ueda). These are the subject of the separate optimal-protocol project. Note that the second law forbids any open-loop protocol from shifting the whole distribution below ΔF.
- The free-energy *profile* $F(\lambda(t))$ from the same pulls (Hummer–Szabo).
- Excluded volume and bending stiffness, with ΔF from thermodynamic integration.

References

[1] C. Jarzynski, Phys. Rev. Lett. **78**, 2690 (1997).

[2] G. E. Crooks, Phys. Rev. E **60**, 2721 (1999).

[3] C. Jarzynski, *Rare events and the convergence of exponentially averaged work values*, Phys. Rev. E **73**, 046105 (2006).

[4] J. Gore, F. Ritort, C. Bustamante, *Bias and error in estimates of equilibrium free-energy differences from nonequilibrium measurements*, PNAS **100**, 12564 (2003).

[5] K. Sekimoto, Prog. Theor. Phys. Suppl. **130**, 17 (1998).

[6] B. Leimkuhler, C. Matthews, Appl. Math. Res. Express **2013**, 34 (2013).

[7] J. Liphardt *et al.*, Science **296**, 1832 (2002).

[8] U. Seifert, Rep. Prog. Phys. **75**, 126001 (2012).